GROUP - 3

University of Waterloo x Brock Solutions: Airport Innovation Challenge

AVIATION



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Aircraft SWAP – Replanner

Aircraft Changed

AC123 · Toronto (YYZ) → Vancouver (YVR) · Scheduled Departure 19:30



192 passengers are checked in

205 checked baggage being prepared

11 cargo items ready

PROBLEM

Late aircraft swap

Existing loading plan INVALID

New plan must be ready before the loading cutoff

Replacement aircraft

- Fewer seats?
- Smaller holds?
- Different balance limits

AFFECTED STAKEHOLDERS

Passengers

Removed by DCS

Baggage

Bags belonging to removed passengers must come off

Cargo

Must be replaced and evaluated for balance limits

Passenger Seat Mapping — Adapting to Aircraft Swaps

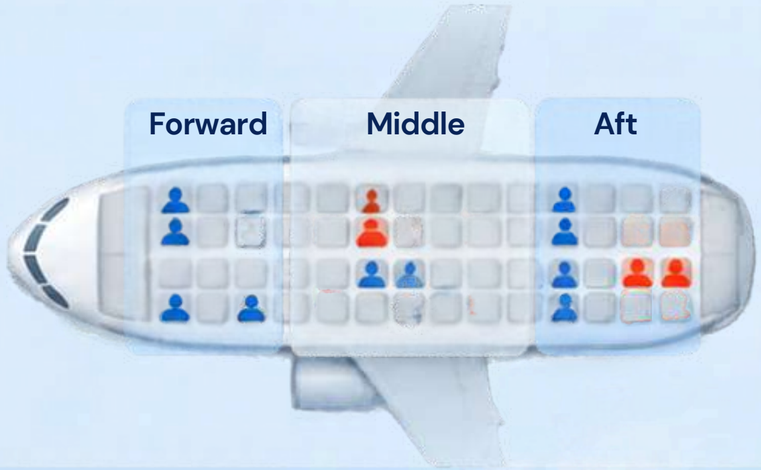
Our Innovation: Aircraft-Adaptive Passenger Mapping
Validates seat assignments and recalculates cabin-zone weights for different replacement aircraft.

Original Aircraft: A321



Aircraft Swap

Replacement Aircraft: A320



Aircraft-Adaptive Mapping

Reads the replacement aircraft's seating configuration.

Invalid Seat Detection

Flags original seat assignments that no longer exist.

Zone Weight Calculation

The Solver Proposes.
The Safety Gate Verifies.

INDEPENDENT VERIFICATION

- Passenger zone weights
- Cargo and baggage accountability
- Hold capacity
- Zero-fuel weight and centre of gravity
- Dangerous-goods restrictions
- Passenger and baggage count consistency

SAFE | NEEDS REVIEW | BLOCKED | UNSAFE

Count inconsistencies can block a plan; infeasible loading is rejected; unresolved warnings require human review.

Clear handling instructions and exception reporting support employees and ramp operators.
Before real-world deployment: approved aircraft data, airline procedure applicable regulations, and appropriate IATA/ICAO guidance.
Decision-support prototype, not certified flight-release software.



Safety Constraints First.
Operational Disruption Second.



Python + PuLP mixed-integer linear programming.

- Maximum zero-fuel weight and centre-of-gravity limits
- Hold weight and volume capacity
- Dangerous-goods placement restrictions
- Must-fly cargo constraints

1. **Minimize baggage left behind**
2. **Preserve high-priority cargo**
3. **Minimize handling movements**
4. **Approach target centre of gravity**

Optional **Lightning Mode** prioritizes fewer handling actions earlier.

Prototype Demonstration.

Working Pipeline

1

Passenger Zone Mapper

Skips passengers removed by DCS, checks seats, sets cabin zones and seasonal weights



2

Bag Offload Tracker

Sorts each bag into fly, pull or left behind, with an ordered pull list



3

Load Replanner

Optimizer assigns holds within weight, volume, balance and DG limits

4

Safety Gate

Recalculates the plan from the manifests and gives it a status

5

Dashboard

Present instructions and results to operators



OUTPUTS: Passenger zone loads, Baggage retrieval lists, Cargo assignments, Operational instructions, Safety review results

✈️ Aircraft Swap Re-planner

Clear-to-fly demonstration · illustrative aircraft profiles · qualified load-controller review required

Summer weights · +0% buffer

Demo setup: inbound bags arrive before the 19:15 cutoff; use summer weights for the clear-to-fly result.

🌐 Combined overview | 🍀 Engine walkthrough

Replacement aircraft · overview only

A320ceo

✅ Safe to load

All limits and counts check out. Ready for the load controller to approve.

✈️ **Aircraft check:** plan calculated for **A320ceo**, the aircraft now assigned to AC123 (was A321ceo). Limits, holds and 162 seats all come from the A320ceo data.

THE FLIGHT AT A GLANCE

AIRCRAFT SWAP	PASSENGERS	CHECKED BAGS	CARGO
A321 → A320	**162 flying**	**172 flying**	**11 items**
AC123 YYZ → YVR · departs 19:30	192 booked · 30 removed by DCS · 162 seats	205 checked · 33 pulled · 0 miss the flight	3,028 kg in total

Passenger weights use Transport Canada standard weights, which already include clothing and carry-on bags. Loading closes 19:15.

1 · THE PROBLEM (BEFORE RE-PLANNING)

HOLDS	WEIGHT IF EVERYTHING IS LOADED	BALANCE (CG) RIGHT NOW
FWD 481 kg over	**61,834 kg**	**16.96 m**
loaded for the bigger aircraft	under the 62,500 kg maximum	inside the safe range

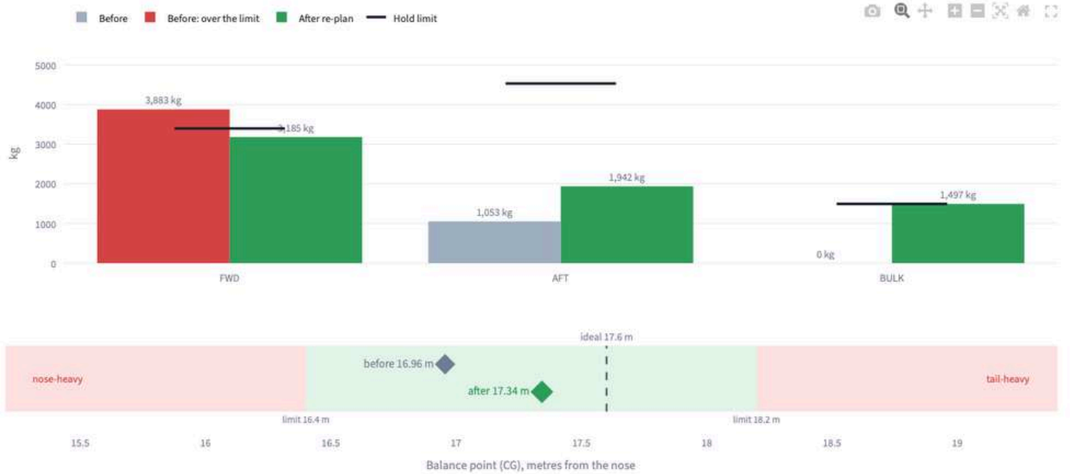
2 · WHAT HAS TO HAPPEN

Extra handling jobs caused by the re-plan: **1** (1 moved, 0 taken out). Everything else stays where it is.

Move Auto parts (698 kg) from FWD to BULK
FWD hold is 481 kg over its limit

Pull 33 bags of passengers removed by the DCS
a bag never flies without its passenger · order in Details

3 · BEFORE AND AFTER



FINAL WEIGHT	FINAL BALANCE (CG)	BAGS
61,834 kg	**17.34 m**	**172 flying**
margin 666 kg (1.1%) under the 62,500 kg maximum	inside the safe range · 0.86 m from the nearest limit	33 pulled · 0 miss the flight

› Loading instructions for the ramp (per hold, pull list, late bags)

4 · LOAD CONTROLLER SIGN-OFF

Load controller name

Approve plan

› What do these words mean?

🌐 Combined overview | 🧩 Engine walkthrough

Choose a pipeline stage

1 · Swap Mapper

Stage 1: Swap Mapper

To decide which retained passengers fit the new aircraft & calculates cabin weight

1 Passenger removed
Passengers on the DCS removal list are skipped, decided by DCS, code takes it as input

2 Check seat
Row and seat letter must be valid, duplicate seats are rejected

3 Assign zone
Each passenger goes to a cabin zone with one balance arm

4 Place lap infants
Placed with a mapped parent or guardian, no seat used

5 Apply weight conditions
Seasonal standard weights, infants are 0 kg unless they exceed 10% of adults

6 Flag misfits
If no valid seat, goes to the “do not fit” list for a DCS decision

Stage 2: Bag Offload Tracker

Classifies every bag and builds the physical pull list.

RULES

#	If the bag is…	Status	Result
1	Location not recognized	● **Unknown**	Held off the flight; plan is BLOCKED
2	Inbound with no ETA or cutoff	● **At risk**	Left behind
3	Inbound, arriving after cutoff	● **At risk**	Left behind, even if the owner was removed
4	Owner removed by DCS	● **To pull**	Added to the pull list
5	Connecting and late	● **At risk**	Left behind
6	Inbound, on time	● **Inbound**	Flies
7	Anything else	● **OK**	Flies

PULL ORDER

Carts → **Holds** → **Inbound**

Within each group: sorted by location, then stack position. Totals are then reconciled against the DCS checked-bag count.

Stage 3: Load Replanner

Places each flying bag and cargo item in one hold, or leaves it off.

HARD LIMITS

Hold weight | Hold volume | Max zero-fuel weight | Forward CG | Aft CG | Forbidden DG classes | Must-fly cargo

GOALS

Standard

1 Fewest bags left → 2 Cargo priority → 3 Fewest moves → 4 CG closest to target

⚡ Lightning mode (avoids moving items already loaded) swaps goals 2 and 3, so it may bump important cargo to save a move

1 Fewest bags left → 2 Fewest moves → 3 Cargo priority → 4 CG closest to target

SOLVER RESULT

OPTIMAL
Best plan found and proven.

FEASIBLE
A plan exists, but optimality was not proven.

INFEASIBLE
No plan meets the limits.

ERROR
The solver could not return a result.

Mixed-integer optimizer (PuLP with CBC) · 10-second limit per stage · cargo priority 1 is most important · every move and bump has a written reason

Stage 4: Safety Gate

Recalculates the final plan from the original manifests, without trusting solver output)

CRITERIA CHECKED

(1) Aircraft limits
- Hold weight and volume
- Zone seats and weight
- Aircraft seat capacity
- Max zero-fuel weight
- Centre of gravity

(2) Counts against DCS
- Passenger accounting
- Bag accounting
- Cargo accounting
- Seat uniqueness

(3) Load rules
- Dangerous goods placement
- Bag-owner matching

DECISION

Status	Trigger	Ready for controller
⛔ BLOCKED	Passenger, seat or bag reconciliation fails	No
🚫 UNSAFE	Any other hard limit fails, or no feasible plan exists	No
🟠 NEEDS REVIEW	CG margin under 0.2 m, MZFW margin under 1%, bags left behind, cargo bumped, or optimality unproven	Yes
✅ SAFE	All checks pass with no warnings	Yes
⚠️ ERROR	The solver could not return a verified result	No

Real Prototype Results:

A320 vs A220.



A320ceo

162	passengers / 162 seats
170	bags assigned to fly
32	bags pulled
3	bags left behind
1	cargo movement
0	cargo items bumped

Zero-fuel weight: 62,204.1 / 62,500 kg

Centre of gravity: 17.338 m



A220-300

160	passengers / 160 seats
168	bags assigned to fly
34	bags pulled
3	bags left behind
3	cargo movements
3	cargo items bumped

Zero-fuel weight: 55,656.8 / 58,000 kg

Centre of gravity: 19.512 m

Different aircraft capacities produce different recovery decisions.



Illustrative simulation values; NEEDS REVIEW is not flight clearance.

Edge Cases Considered

Area	Situation	What happens
Passengers	Removed by DCS	Skipped, and their bags are pulled
	Invalid row or seat, or row not on the new aircraft	“Do not fit” list, plan BLOCKED
	Duplicate seat, or over seat capacity	BLOCKED
	Lap infant with a mapped parent	Placed with the parent, uses no seat
	Lap infant with no parent	“Do not fit” list, plan BLOCKED
	Removed passenger appears in the plan	BLOCKED
	Unknown passenger category	Stopped with an error by the mapper, UNSAFE at the gate
Bags	Unknown location	Held off the flight, BLOCKED
	Inbound with no ETA	At risk, left behind
	Inbound after cutoff	Left behind, even if the owner was removed
	ETA after midnight, or with seconds	Handled correctly
	Bag loaded without its owner	Rejected by the gate
Cargo and holds	Hold over its limit	Items moved to another hold
	Not enough room	Cargo bumped, lowest priority first
	Must-fly cargo can’t fit	UNSAFE, no plan offered
	Item in a hold the new aircraft lacks	Recorded as a move
Data	Text flags such as "false"	Read correctly, invalid values raise an error
Counts	DCS count mismatch	BLOCKED
	Mismatch and no feasible plan	BLOCKED
Solver	Times out with a usable plan	FEASIBLE, goes to review
	Crashes or returns nothing	ERROR, no plan offered

Practical Deployment, Cost and ROI.

Cost

Built with Python, Streamlit, PuLP/CBC, and structured JSON data; no new airport hardware is required for the prototype.

Real deployment costs include airline integration, approved aircraft datasets, validation, training, support, and maintenance.



Potential Return

- Less manual conciliation
- Clearer loading and offloading decisions
- Fewer unnecessary cargo movements
- Better visibility into safety exceptions
- Potentially faster disruption recovery



Practicality

Petontial integration with airline Departure Control Systems and baggage/cargo systems follows further engineering and validation.



Reliability

Input validation, independent verification, failure states, human review, and future recovery procedures.



Potential benefits, not measured financial savings; no costs or performance statistics are claimed.

Faster Recovery. Clearer Decisions. Safety First.

1. **Connected Recovery**

Passenger loading, baggage conciliation, and cargo optimization work together.

2. **Actionable Decisions**

Ground teams receive understandable loading, movement, and offloading instructions.

3. **Independent Safety Verification**

Unsafe or inconsistent plans are identified before operational acceptance.

Future development:
Approved aircraft weight-and-balance data, operational DCS and baggage-system integration, broader disruption testing; operator usability validation; real-world performance and cost benchmarking.



https://github.com/nihellohaooooo/F26-Airport-Innovation-Challenge-Group3

AID Statement: Artificial Intelligence Tool: Claude Opus 5.5, Codex 6 Luna, ChatGPT.

Q&A


THANK YOU